Renan Souza, Ph.D.

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Summary

Research staff scientist focusing on workflow provenance, data integration, and agentic AI workflows for science. I develop methods and systems that improve transparency, reliability, and reproducibility in end-to-end workflows, including the AI lifecycle. I also build and deploy open-source software and collaborate nationally and internationally across scientific domains.

Areas of Expertise

Workflow provenance and reproducibility in large-scale science ◆ Provenance-aware data management for the AI lifecycle ◆ Agentic AI workflows with traceability and reliability ◆ Edge-Cloud-HPC workflow integration and federated operations ◆ Scalable workflow analytics and human-in-the-loop steering

Education

Federal University of Rio de Janeiro, Brazil - Rio de Janeiro, Brazil

Ph.D. in Computer Science | Sep 2015 — Dec 2019 | COPPE/Data and Knowledge Engineering
Thesis: Supporting User Steering in Large-scale Workflows with Provenance Data
Supervisor: Marta Mattoso and Patrick Valduriez.

M.Sc. in Computer Science | Jan 2013 — Jul 2015 | COPPE/Data and Knowledge Engineering
Thesis: Controlling the Parallel Execution of Workflows Relying on a Distributed Database
Supervisor: Marta Mattoso

B.Sc. in Computer Science | Jan 2009 — Dec 2012
Thesis: Linked Open Data Publication Strategies: An Application in Network Performance Data
Supervisor: Maria Luiza Machado Campos

International experience:

Visiting Ph.D. Student - Inria/Univ. Montpellier, France | Jan 2019 — Mar 2019 (Ph.D.)
Computer Science exchange student - Missouri State University, U.S. | Jun 2011 — Jun 2012 (B.Sc.)

Experience

Oak Ridge National Laboratory **Oct 2022 — Present**
Staff Scientist & Sr. Software Engineer, HPC Workflows, Data & AI **Knoxville, USA**

- Led R&D on workflow provenance and observability for AI-driven science, focusing on transparency, reliability, and reproducibility in end-to-end workflows.
- Developed provenance models and system mechanisms to connect user intent, agent decisions, workflow executions, and downstream results in unified traces.
- Validated methods through real deployments spanning Edge-Cloud-HPC environments and interactive workflows that require low-latency, auditable responses.
- Published and presented results in workflow and eScience venues, and drove community engagement through tutorials and reference architectures.

IBM Research **Apr 2015 — Oct 2022**
Staff Scientist & Sr. Software Engineer, Cloud, Data & AI **Rio de Janeiro, Brazil**

- Conducted applied R&D in data management and AI systems, producing peer-reviewed outputs and patented innovations.
- Explored scalable data services and governance-aware approaches that support AI lifecycle requirements in enterprise contexts.
- Collaborated internationally across research and engineering teams to validate ideas in real deployments and user scenarios.

SLAC National Accelerator Laboratory, Stanford University **May 2013 — Dec 2014**
Research Software Engineering Intern **Menlo Park, USA**

- Applied semantic web and scalable data management methods to publish structured measurement data for broad community use.

Federal University of Rio de Janeiro **Jan 2010 — Sep 2014**
Software Engineer (Intern → Engineer) **Rio de Janeiro, Brazil**

- Built applied semantic web and linked data solutions, grounding research ideas in real systems and user needs.
- Early applied work on integrating heterogeneous data sources for analytics and reporting.

Petrobras **May 2007 — May 2008**
IT Intern **Rio de Janeiro, Brazil**

- Early industry experience in software delivery and operations.

Selected on-going Projects

American Science Cloud (AmSC)

AmSC integrates DOE computing and experimental facilities, high-performance networks, and data resources into a secure, federated environment for AI-ready datasets and model services. I design and implement multi-agent workflow frameworks within ORNL use cases and align them with platform APIs for cross-facility interoperability.

Orchestrated Platform for Autonomous Laboratories (OPAL)

OPAL seeks to create a network of autonomous laboratories that can learn and adapt by combining AI, robotics, and automated experimentation. I lead agentic AI systems that translate expert intent into HPC-scale execution and multimodal analytics, enabling human-in-the-loop discovery workflows.

Advanced Manufacturing into Leadership-class Supercomputers via AI Agents

A cross-facility system integrating a natural language interface, a multi-agent decision framework, programmable facility APIs, and provenance-aware infrastructure for adaptive and reproducible workflows. I led the multi-agent communication layer and dynamic steering mechanisms that connect decisions to actions across facilities.

Flowcept

Flowcept captures runtime provenance with low overhead and links tasks, lineage, telemetry, and AI-agent interactions into end-to-end traces for accountability and reproducibility. I created and lead the platform, which underpins multiple DOE cross-facility initiatives and current work on agentic provenance.

Technical Knowledge

Programming Languages: Python, Java, C, C++, C#, Shell, NodeJS, Scala, Lua
Data Science/ML: PyTorch, MLFlow, Airflow, Pandas, Polars, Jupyter, Matplotlib, Plotly
Agentic AI: MCP, CrewAI, LangChain, Streamlit, Chainlit, RAG, LLM-based orchestration
Big Data & Streaming Platforms: Apache Spark, Dask, Kafka, Redis, RabbitMQ
Parallel & Distributed Programming: MPI, OpenMP, CUDA, PubSub
Cloud, HPC, DevOps: Kubernetes, OpenShift, Docker, Slurm, LSF, PBS; CI/CD GitHub Actions, Jenkins, Travis; Grafana
Databases & Knowledge Graphs: PostgreSQL/PostGIS, MySQL, MongoDB, Elasticsearch, HBase, Hive, Redis, LMDB, Polystores; AllegroGraph, Jena, Virtuoso, RDF, SPARQL, OWL

Selected Publications

Complete list of publications and patents is in the end of this document.

- **R. Souza**, T. J. Skluzacek, S. R. Wilkinson, M. Ziatdinov, and R. F. da Silva, "Towards Lightweight Data Integration using Multi-workflow Provenance and Data Observability," in IEEE International Conference on e-Science, 2023, doi: 10.1109/e-Science58273.2023.10254822.

- **R. Souza**, L. G. Azevedo, V. Lourenço, E. Soares, R. Thiago, et al., "Workflow Provenance in the Lifecycle of Scientific Machine Learning," Concurrency and Computation: Practice and Experience, 2021, doi: 10.1002/cpe.6544.
- **R. Souza**, A. Gueroudji, S. DeWitt, D. Rosendo, T. Ghosal, et al., "PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows," in IEEE International Conference on e-Science, 2025.
- **R. Souza**, S. Caino-Lores, M. Coletti, T. J. Skluzacek, A. Costan, et al., "Workflow Provenance in the Computing Continuum for Responsible, Trustworthy, and Energy-Efficient AI," in IEEE International Conference on e-Science, 2024, doi: https://doi.org/10.1109/e-Science62913.2024.10678731.

Grants and Awards

- ORNL performance award for top performers (2025)
- LDRD-funded project leadership, managed near $2M and led 10+ researchers and developers (2023—2025)
- 2nd IBM Patent Plateau for high-impact software and data innovations (8+ USPTO patents) (2021)
- SBBD Honored Mention for the Best Ph.D. Thesis Award (provenance and workflow/data management) (2021)
- 1st IBM Patent Plateau for software and systems contributions (4+ USPTO patents) (2020)
- Early Academic Honors: SBBD Best M.Sc. Thesis and CAPES International Science Grant (2012—2017)

Teaching and Supervisions

Courses

- **Databases Laboratory** (UFRJ, 2017). Teacher assistant to Prof. Marta Mattoso
- **Logics for Computer Science** (UFRJ, 2012—2013). Teacher assistant to Prof. Mario Benevides

Supervisions

- Pedro Paiva Miranda, *A Mechanism for Fault Tolerance in Parallel Executions of Workflows supported by a Database* (undergraduate, 2015)
- Rachel Gonçalves de Castro, *Publication of Workflow Provenance Data in the Semantic Web* (undergraduate, 2015)

Scientific Community Service

- IEEE Transactions on Parallel and Distributed Systems - Reviewer
- Future Generation Computer Systems - Reviewer
- Concurrency Computation Practice and Experience - Reviewer
- Journal of Parallel and Distributed Computing - Reviewer
- The Very Large Databases (VLDB) Journal - Reviewer
- IEEE Transactions on Big Data - Reviewer
- Journal of Cloud Computing - Reviewer
- Computer Physics Communications - Reviewer
- Discover Data - Reviewer
- Frontiers in High Performance Computing - Reviwer and Editorial Board
- International Workshop on AI Principles in Science Communication (AISC) - PC
- International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD'25) - PC
- Workflows in Distributed Environments (WiDE'24) - PC
- IEEE/ACM Supercomputing (SC'24) - PC
- IEEE International Conference on e-Science (eScience'23) - Session Chair, PC
- Workflows in Support of Large-Scale Science (WORKS'20, 21, 23, 24, 25) - PC
- Brazilian Workshop on Database and Artificial Intelligence Integration - PC
- Brazilian Symposium on Databases (SBBD'20, 23, 24, 25, 26) - PC, Session Chair
- Brazilian e-Science (BreSci'26) - PC
- Innovation Summit on Information Systems (at SBSI'19,20) - PC

Badges and Certifications

- Machine Learning Specialist Professional (2022). Exploratory Data Analysis, Regression, Classification, Deep Learning, Reinforcement Learning, Unsupervised Learning, Time Series and Survival Analysis, AI Ethics and Explainability
- Trustworthy AI and AI Ethics (2022). Trustworthy AI foundations and applied AI ethics
- Enterprise Design Thinking Practitioner (2022)
- LinkedIn Skill Assessment: Python, MySQL, Linux, T-SQL, NoSQL

Languages

- English: Full professional proficiency
- Portuguese: Native
- Spanish: Reading fluent; speaking/listening intermediate

Talks and Events

- **Workflows Community Initiative** (2025) | Remote
- Oral presentation: For Provenance and with Provenance: The Role of Provenance Data in Agentic Workflows
- **IEEE/ACM Supercomputing (SC)** (2025) | St. Louis, U.S.A.
- Oral presentation: LLM Agents for Interactive Workflow Provenance: Reference Architecture and Evaluation Methodology
- **IEEE International Conference on e-Science** (2025) | Chicago, U.S.A
- Tutorial: Large-scale Workflow Provenance Data Management in the AI Lifecycle using Flowcept
- Oral presentation: PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows
- **IEEE International Conference on e-Science** (2024) | Osaka, Japan (Virtual)
- Oral presentation: Workflow Provenance in the Computing Continuum for Responsible, Trustworthy, and Energy-Efficient AI
- **IEEE/ACM Supercomputing (SC)** (2024) | Atlanta, GA
- Oral presentation: Integrating Evolutionary Algorithms with Distributed Deep Learning for Optimizing Hyperparameters on HPC Systems
- **IEEE/ACM Supercomputing (SC)** (2023) | Denver, CO
- **IEEE International Conference on e-Science** (2023) | Limassol, Cyprus
- Oral presentation: Towards Lightweight Data Integration using Multi-workflow Provenance and Data Observability
- **Brazilian Symposium on Databases (SBBD)** (2021) | Rio de Janeiro, RJ (virtual)
- Oral presentation: User Steering Support in Large-Scale Workflows
- **Federal Fluminense University (UFF) Computer Science Seminars** (2021) | Rio de Janeiro, RJ (virtual)
- Invited talk (Portuguese): A Knowledge-centric Approach to Support Large-scale AI Systems
- **SIAM Conference on Computational Science and Engineering** (2021) | Forth Worth, TX (virtual)
- Invited talk, Highlighted by the SIAM press: AI4Seismic: An AI-Driven Platform to Accelerate Geological Discoveries
- Oral presentation: Workflow Provenance in the Lifecycle of Scientific Machine Learning
- **ACM International Conference on Management of Data (SIGMOD)** (2020) | Portland, OR (virtual)
- **Brazilian Symposium on Databases (SBBD)** (2020) | Rio (virtual)
- **High-Performance Data Science workshop** (2020) | Rio (virtual)
- **Computational Science and Engineering Seminar at COPPE/UFRJ** (2020) | Rio (virtual)
- Invited talk: Workflow Provenance in the Lifecycle of Scientific Machine Learning
- **Open Subsurface Data Universe Development Workshop** (2020) | Houston, TX
- **Open Subsurface Data Universe Development Workshop** (2019) | Houston, TX
- **IEEE/ACM Supercomputing (SC)** (2019) | Denver, CO
- **Scientific Data Analysis using Data-intensive Scalable Computing Workshop** (2019) | Rio de Janeiro, Brazil

- Invited talk: Provenance Data in the Machine Learning Lifecycle in Computational Science and Engineering
- **Open Subsurface Data Universe F2F Meeting** (2019) **|** Houston, TX
- **IEEE International Conference on e-Science** (2019) **|** San Diego, CA
- Oral presentation: Efficient Runtime Capture of Multiworkflow Data using Provenance
- **Inria Talks** (2019) **|** Montpellier, France
- Invited talk: Providing Online Data Analytical Support for Humans in the Loop of Computational Science and Engineering Applications
- **IBM Regional Technical Exchange** (2019) **|** Rio de Janeiro, Brazil
- **Provenance Week** (2018) **|** London, UK
- **International Conference on Very Large Databases (VLDB)** (2018) **|** Rio de Janeiro, Brazil
- **Brazilian Syposium on Databases (SBBD)** (2018) **|** Rio de Janeiro, Brazil
- **Brazilian Syposium on Databases (SBBD)** (2017) **|** Uberlandia, Brazil
- Oral presentation: Spark Scalability Analysis in a Scientific Workflow
- Oral presentation: Controlling the Parallel Execution of Workflows Relying on a Distributed Database
- **Federal University of Uberlandia, Brazil** (2017) **|** Uberlandia, Brazil
- Invited talk: Kubernetes
- **Smart City Cloud Hackathon OpenStack Rio** (2017) **|** Rio de Janeiro, Brazil
- **Computer Science Week at UFRJ** (2017) **|** Rio de Janeiro, Brazil
- Oral presentation: Kubernetes
- **Brazilian Conference on Artificial Intelligence (BRACIS)** (2017) **|** Recife, Brazil
- Tutorial: Graph Analytics with Spark
- **IEEE/ACM Supercomputing (SC)** (2016) **|** Salt Lake City, UT
- **ASE BigData/SocialCom/CyberSecurity** (2014) **|** Stanford University, Menlo Park, CA
- poster presentation: Linked open data publication strategies: Application in networking performance measurement data

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All Publications and Patents

Journal Articles

[J1] M. Dorier, A. Gueroudji, V. Hayot-Sasson, H. Nguyen, S. Ockerman, **R. Souza**, T. Bicer, H. Pan, P. Carns, K. Chard, and others, "Toward a Persistent Event-Streaming System for High-Performance Computing Applications," *Frontiers in High Performance Computing*, 2025. DOI: 10.3389/fhpcp.2025.1638203

[J2] D. Bard, K. Chard, S.d. Witt, I.T. Foster, C. Goble, W. Godoy, J. Gustafsson, U. Haus, S. Hudson, L. Los, **R. Souza**, and others, "Workflows Community Summit 2024: Future Trends and Challenges in Scientific Workflows," *Distributed, Parallel, and Cluster Computing (cs.DC)*, 2024. DOI: https://doi.org/10.48550/arXiv.2410.14943

[J3] R.F.d. Silva, R.M. Badia, V. Bala, D. Bard, P. Bremer, I. Buckley, S. Caino-Lores, K. Chard, C. Goble, S. Jha, ..., **R. Souza**, and e. al., "Workflows Community Summit 2022: A Roadmap Revolution," *arXiv preprint Distributed, Parallel, and Cluster Computing (cs.DC)*, 2023. DOI: 10.48550/arXiv.2304.00019

[J4] L.G. Azevedo, **R. Souza**, E.F.d.S. Soares, R.M. Thiago, J.C.C. Tesolin, A.C. Oliveira, and M.F. Moreno, "A Polystore Architecture Using Knowledge Graphs to Support Queries on Heterogeneous Data Stores," *arXiv preprint Databases (cs.DB)*, 2023. DOI: 10.48550/arXiv.2308.03584

[J5] **R. Souza**, V. Silva, A.A.B. Lima, D. Oliveira, P. Valduriez, and M. Mattoso, "Distributed In-memory Data Management for Workflow Executions," *PeerJ Computer Science*, 2021. DOI: 10.7717/peerj-cs.527

[J6] R.F.d. Silva, H. Casanova, K. Chard, ..., **R. Souza**, and e. al., "Workflows Community Summit: Advancing the State-of-the-art of Scientific Workflows Management Systems Research and Development," *arXiv preprint Distributed, Parallel, and Cluster Computing (cs.DC)*, 2021.

[J7] **R. Souza**, L.G. Azevedo, V. Lourenço, E. Soares, R. Thiago, R. Brandão, D. Civitarese, E.V. Brazil, M. Moreno, P. Valduriez, M. Mattoso, R. Cerqueira, and M.A.S. Netto, "Workflow Provenance in the Lifecycle of Scientific Machine Learning," *Concurrency and Computation: Practice and Experience*, 2021.

[J8] L.G. Azevedo, **R. Souza**, R. Brandão, V.N. Lourenço, M. Costalonga, M.d. Machado, M. Moreno, and R. Cerqueira, "Adding Hyperknowledge-enabled data lineage to a machine learning workflow management system for oil and gas," *First Break*, 2020. DOI: 10.3997/1365-2397.fb2020055

[J9] **R. Souza**, V. Silva, J.J. Camata, A.L.G.A. Coutinho, P. Valduriez, and M. Mattoso, "Keeping Track of User Steering Actions in Dynamic Workflows," *Future Generation Computer Systems*, 2019. DOI: 10.1016/j.future.2019.05.011

[J10] V. Silva, L. Neves, **R. Souza**, A.L.G.A. Coutinho, D.d. Oliveira, and M. Mattoso, "Adding Domain Data to Code Profiling Tools to Debug Workflow Parallel Execution," *Future Generation Computer Systems*, 2018. DOI: 10.1016/j.future.2018.05.078

[J11] M.G.d. Bayser, P. Cavalin, **R. Souza**, A. Braz, H. Candello, C. Pinhanez, and J. Briot, "A Hybrid Architecture for Multi-party Conversational Systems," *arXiv preprint Computation and Language (cs.CL)*, 2017.

[J12] **R. Souza**, V. Silva, A.L.G.A. Coutinho, P. Valduriez, and M. Mattoso, "Data Reduction in Scientific Workflows Using Provenance Monitoring and User Steering," *Future Generation Computer Systems*, 2017. DOI: 10.1016/j.future.2017.11.028

Conference and Workshop Papers

[C1] **R. Souza**, A. Gueroudji, S. DeWitt, D. Rosendo, T. Ghosal, R. Ross, P. Balaprakash, and R.F.d. Silva, "PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows," *IEEE International Conference on e-Science*, 2025.

[C2] **R. Souza**, T. Poteet, B. Etz, D. Rosendo, A. Gueroudji, W. Shin, P. Balaprakash, and R.F.d. Silva, "LLM Agents for Interactive Workflow Provenance: Reference Architecture and Evaluation Methodology," *Workflows in Support of Large-Scale Science (WORKS) co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2025. DOI: 10.1145/3731599.3767582

[C3] W. Shin, **R. Souza**, D. Rosendo, F. Suter, F. Wang, P. Balaprakash, and R.F.d. Silva, "The (R) evolution of Scientific Workflows in the Agentic AI Era: Towards Autonomous Science," *Workflows in Support of Large-Scale Science (WORKS) co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2025.

[C4] D. Rosendo, S. DeWitt, **R. Souza**, P. Austria, T. Ghosal, M. McDonnell, R. Miller, T.J. Skluzacek, J. Haley, B. Turcksin, and others, "AI Agents for Enabling Autonomous Experiments at ORNL's HPC and Manufacturing User Facilities," *Extreme-Scale Experiment-in-the-Loop Computing (XLOOP) co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2025. DOI: 10.1145/3731599.3767592

[C5] B. Etz, S. Oral, R.F.D. Silva, R. Adamson, A. Alnajjar, T. Beck, A. Barker, M. Brim, P. Bryant, **R. Souza**, and others, "OLCF's Advanced Computing Ecosystem (ACE): FY25 Update for Ongoing Efforts," **, 2025. DOI: 10.2172/3006499

[C6] A. Gueroudji, T. Mallick, **R. Souza**, R.F.d. Silva, R. Ross, M. Dorier, P. Carns, K. Chard, and I. Foster, "ControlA: Agentic Workflow Control Mechanisms for Reliable Science," *IEEE International Conference on e-Science*, 2025.

[C7] **R. Souza**, S. Caino-Lores, M. Coletti, T.J. Skluzacek, A. Costan, F. Suter, M. Mattoso, and R.F.d. Silva, "Workflow Provenance in the Computing Continuum for Responsible, Trustworthy, and Energy-Efficient AI," *IEEE International Conference on e-Science*, 2024. DOI: https://doi.org/10.1109/e-Science62913.2024.10678731

[C8] T.J. Skluzacek, **R. Souza**, M. Coletti, F. Suter, and R.F.d. Silva, "Towards Cross-Facility Workflows Orchestration through Distributed Automation," *Practice and Experience in Advanced Research Computing (PEARC 24)*, 2024. DOI: 10.1145/3626203.3670606

[C9] R.F.d. Silva, W. Shin, F. Suter, A. Gainaru, **R. Souza**, D. Dietz, and S. Jha, "Eco-Driven AI-HPC: Optimizing Energy Efficiency in Distributed Scientific Workflows," *Energy-Efficient Computing for Science Workshop*, 2024.

[C10] R.F.d. Silva, K. Maheshwari, T. Skluzacek, **R. Souza**, and S. Wilkinson, "Advancing Computational Earth Sciences: Innovations and Challenges in Scientific HPC Workflows," *European Geosciences Union (EGU)*, 2024.

[C11] M. Coletti, **R. Souza**, T.J. Skluzacek, F. Suter, and R.F.d. Silva, "Integrating Evolutionary Algorithms with Distributed Deep Learning for Optimizing Hyperparameters on HPC System," *Workflows in Support of Large-Scale Science (WORKS) workshop co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2024.

[C12] L.G. Azevedo, **R. Souza**, E. Soares, R.M. Thiago, J.C.C. Tesolin, A.C.C.M. Oliveira, and M.F. Moreno, "HKPoly: A Polystore Architecture to Support Data Linkage and Queries on Distributed and Heterogeneous Data," *Proceedings of the 20th Brazilian Symposium on Information Systems (SBSI)*, 2024. DOI:

10.1145/3658271.3658322

[C13] **R. Souza**, T.J. Skluzacek, S.R. Wilkinson, M. Ziatdinov, and R.F.d. Silva, "Towards Lightweight Data Integration using Multi-workflow Provenance and Data Observability," *IEEE International Conference on e-Science*, 2023. DOI: 10.1109/e-Science58273.2023.10254822

[C14] D. Rosendo, M. Mattoso, A. Costan, **R. Souza**, D. Pina, P. Valduriez, and G. Antoniu, "ProvLight: Efficient Workflow Provenance Capture on the Edge-to-Cloud Continuum," *IEEE International Conference on Cluster Computing*, 2023. DOI: 10.1109/CLUSTER52292.2023.00026

[C15] **R. Souza**, "User Steering Support in Large-scale Workflows," *PhD Thesis Contest: Brazilian Symposium on Databases (SBBD)*, 2021.

[C16] E. Soares, **R. Souza**, R. Thiago, M. Machado, and L. Azevedo, "A Recommender for Choosing Data Systems based on Application Profiling and Benchmarking," *Brazilian Symposium on Databases (SBBD)*, 2021.

[C17] R.L. Cunha, L.V. Real, **R. Souza**, B. Silva, and M.A. Netto, "Context-aware Execution Migration Tool for Data Science Jupyter Notebooks on Hybrid Clouds," *IEEE International Conference on e-Science*, 2021. DOI: 10.1109/eScience51609.2021.00013

[C18] L. Azevedo, **R. Souza**, E. Soares, R. Thiago, A. Oliveira, and M. Moreno, "Supporting Polystore Queries using Provenance in a Hyperknowledge Graph," *International Semantic Web Conference (ISWC)*, 2021.

[C19] **R. Souza**, J. Camata, M. Mattoso, and A. Coutinho, "Runtime Steering of Parallel CFD Simulations," *International Conference on Parallel Computational Fluid Dynamics*, 2020.

[C20] R. Thiago, **R. Souza**, L. Azevedo, E. Soares, R. Santos, W. Santos, M.D. Bayser, M. Cardoso, M. Moreno, and R. Cerqueira, "Managing Data Lineage of O\&G Machine Learning Models: The Sweet Spot for Shale Use Case," *European Association of Geoscientists and Engineers (EAGE) Digitalization Conference and Exhibition*, 2020. DOI: 10.3997/2214-4609.202032075

[C21] **R. Souza**, A. Codas, J.A.N. Junior, M.P. Quinones, L. Azevedo, R. Thiago, E. Soares, M. Cardoso, and L. Martins, "Supporting the Training of Physics Informed Neural Networks for Seismic Inversion Using Provenance," *American Association of Petroleum Geologists Annual Convention and Exhibition (AAPG)*, 2020.

[C22] R. Brandão, V. Lourenço, M. Machado, L. Azevedo, M. Cardoso, **R. Souza**, G. Lima, R. Cerqueira, and M. Moreno, "A Knowledge-Based Approach for Structuring Cyclic Workflows," *International Semantic Web Conference (ISWC)*, 2020.

[C23] R. Brandão, V. Lourenço, M. Machado, L. Azevedo, M. Cardoso, **R. Souza**, G. Lima, R. Cerqueira, and M. Moreno, "Cycle Orchestrator: A Knowledge-Based Approach for Structuring Cyclic ML Pipelines in the O\&G Industry," *International Semantic Web Conference (ISWC)*, 2020.

[C24] L. Azevedo, **R. Souza**, E. Soares, and M. Moreno, "Modern Federated Databases: an Overview," *International Conference on Enterprise Information Systems (ICEIS)*, 2020.

[C25] L. Azevedo, **R. Souza**, R. Thiago, E. Soares, and M. Moreno, "Experiencing ProvLake to Manage the Data Lineage of AI Workflows," *Innovation Summit on Information Systems (EISI) in Brazilian Symposium in Information Systems (SBSI)*, 2020.

[C26] **R. Souza**, L. Azevedo, V. Lourenço, E. Soares, R. Thiago, R. Brandão, D. Civitarese, E.V. Brazil, M. Moreno, P. Valduriez, M. Mattoso, R. Cerqueira, and M.A.S. Netto, "Provenance Data in the Machine Learning Lifecycle in Computational Science and Engineering," *Workflows in Support of Large-Scale Science (WORKS) co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2019. DOI: 10.1109/WORKS49585.2019.00006

[C27] **R. Souza**, E.V. Brazil, L. Azevedo, R. Ferreira, D. Chevitarese, E. Soares, R. Thiago, M. Nery, V. Torres, and R. Cerqueira, "Managing Data Traceability in the Data Lifecycle for Deep Learning Applied to Seismic Data," *American Association of Petroleum Geologists Annual Convention and Exhibition (AAPG)*, 2019.

[C28] **R. Souza**, L. Azevedo, R. Thiago, E. Soares, M. Nery, M. Netto, E.V. Brazil, R. Cerqueira, P. Valduriez, and M. Mattoso, "Efficient Runtime Capture of Multiworkflow Data Using Provenance," *IEEE International Conference on e-Science*, 2019. DOI: 10.1109/eScience.2019.00047

[C29] P. Valduriez, M. Mattoso, R. Akbarinia, H. Borges, J. Camata, A.L.G.A. Coutinho, D. Gaspar, N. Lemus, J. Liu, H. Lustosa, F. Masseglia, F.N.D. Silva, V. Silva, **R. Souza**, K. Ocaña, E. Ogasawara, D. Oliveira, E. Pacitti, F. Porto, and D. Shasha, "Scientific Data Analysis Using Data-Intensive Scalable Computing: the SciDISC Project," *LADaS: Latin America Data Science Workshop*, 2018.

[C30] **R. Souza**, L. Neves, L. Azeredo, R. Luiz, E. Tady, P. Cavalin, and M. Mattoso, "Towards a human-in-the-loop library for tracking hyperparameter tuning in deep learning development," *Latin American Data Science (LaDaS) workshop co-located with the Very Large Database (VLDB) conference*, 2018.

[C31] M.G.d. Bayser, C. Pinhanez, H. Candello, M. Affonso, M.P. Vasconcelos, M.A. Guerra, P. Cavalin, and **R. Souza**, "Ravel: A MAS orchestration platform for Human-Chatbots Conversations," *International Workshop on Engineering Multi-Agent Systems (EMAS@AAMAS 2018)*, 2018.

[C32] **R. Souza** and M. Mattoso, "Provenance of Dynamic Adaptations in User-Steered Dataflows," *Provenance and Annotation of Data and Processes - International Provenance and Annotation Workshop (IPAW)*, 2018. DOI: 10.1007/978-3-319-98379-0_2

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